

Naive Bayes

Importing the libraries

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

Importing the dataset

```
In [6]: df= pd.read_csv(r"C:\Users\sande\OneDrive\Desktop\Naresh IT\data science and ai\Mac
df
```

```
Out[6]:
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

```
In [7]: X = df.iloc[:, [2, 3]].values
y = df.iloc[:, -1].values
```

Splitting the dataset into the Training set and Test set

```
In [9]: from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.20, random_
```

Feature Scaling

```
In [10]: from sklearn.preprocessing import Normalizer
sc = Normalizer()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

Training the Naive Bayes model on the Training set

```
In [11]: from sklearn.naive_bayes import MultinomialNB
classifier = MultinomialNB()
classifier.fit(X_train, y_train)
```

Out[11]:

▼ MultinomialNB ⓘ ?

▼ Parameters

alpha	1.0
force_alpha	True
fit_prior	True
class_prior	None

Predicting the Test set results

```
In [12]: y_pred = classifier.predict(X_test)
```

Making the Confusion Matrix

```
In [13]: from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_test, y_pred)
print(cm)
```

```
[[58  0]
 [22  0]]
```

```
In [14]: from sklearn.metrics import accuracy_score
ac = accuracy_score(y_test, y_pred)
print(ac)
```

```
0.725
```

```
In [15]: bias = classifier.score(X_train, y_train)
bias
```

Out[15]: 0.621875

In []: